

BSDS (2024 Semester V): Regression Techniques

Quiz 1

NAME :

ROLL :

Time: 60 minutes

Total marks: 20

Answer all questions. Give explanation to each step used to solve the problems.

1. Let $(Y_1, Y_2, Y_3)^\top \sim N_3((\mu_1, \mu_2, \mu_3)^\top, \Sigma)$ where $\Sigma = (1 - \rho)I_3 + \rho \underline{1}\underline{1}^\top$.

- (a) Define $\underline{Z} = (Y_1 - Y_2, Y_2 - Y_3, Y_1 + Y_2 + Y_3)^\top$. Find the distribution of $\underline{Z}$.
- (b) Hence, or otherwise find conditional distribution of $(Y_1 - Y_2, Y_2 - Y_3)^\top$ given $\bar{Y}$.
- (c) Define $\underline{W} = \underline{Y} + \underline{a}$ where $\underline{a} \sim \mathbb{R}^3$. Find the conditional distribution of $\underline{W}$ given $\bar{Y}$.

[4 + 2 + 6]

2. Let $X_i \stackrel{\text{ind.}}{\sim} \text{Gamma}(\alpha_i, 1)$ for $i = 1, \dots, p$. Define

$$Y_p = X_1 + \dots + X_p, \quad Y_i = X_i/Y_p \quad \text{for } i = 1, \dots, p-1.$$

(a) Using the transformation technique, find the joint pdf of $\underline{Y}_p = [Y_1 \dots Y_p]^\top$.

[Hint: 1. Let $X \sim \text{Gamma}(\alpha, 1)$, then $f_X(x) = \{\Gamma(\alpha)\}^{-1} x^{\alpha-1} \exp\{-x\}$, $x > 0$, $\alpha > 0$.

2. While finding the Jacobian, you may use the fact

$$\det \left(\begin{bmatrix} A & B \\ C & D \end{bmatrix} \right) = \det(A) \times \det(D - CA^{-1}B).$$

(b) Hence, find the joint pdf of $[Y_1 \dots Y_{p-1}]^\top$ and the marginal distribution of Y_p . [5 + 3]

Rough work